

Construction of metallo-triangles with *cis*-TPE motifs and fluorescence properties†Cite this: *Chem. Commun.*, 2022, 58, 13767Received 25th October 2022,
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Two metallo-triangles, SA and SB, with *cis*-TPE motifs were constructed, and their fluorescence properties were explored. Compared with the dilute solution, both triangles SA and SB exhibited significant AIE behavior in the aggregated states. Moreover, the shorter version SA showed higher quantum yields than SB in the aggregated states.

In the past decades, aggregation-induced emission (AIE) has received considerable attention in light-emitting materials. Opposed to the aggregation-caused quenching (ACQ) effect of conventional organic fluorophores, AIE is a phenomenon by which aggregates exhibit strong emission by restricting the intramolecular motion of twisted luminophores, thus reducing non-radiative decay.^{1,2} As classical representative AIE fluorophores with simple molecular structures and special geometry, tetraphenylethylene (TPE) and its derivatives have been widely investigated.^{3,4} Due to the rotation of four benzene rings and the twist of the ethylenic C=C bond in the dilute solution, the fluorescence is quenched.⁵ However, in the aggregated state, it has strong fluorescence emission owing to the restriction of its intramolecular motion, showing AIE behaviour.

Coordination-driven self-assembly as an effective bottom-up approach has been widely used in building various sophisticated supramolecular coordination complexes (SCCs) with well-defined sizes, shapes, geometries, and diverse functionality because of definite directionality and high predictability of

coordination bonds.^{6–9} Among various and diverse organic ligands, 2,2',6',2''-terpyridine has been extensively used as the coordination motif thanks to the prominent chelating ability and outstanding coordination ability with diverse metal ions.^{10–13} With the aim of improving their functionality, SCCs incorporated with chromophores have evolved to be one of the most attractive topics on account of their broad applications in light-emitting diodes,¹⁴ sensors,¹⁵ photoelectric devices,¹⁶ bioimaging,^{17,18} etc. Among various chromophores, as classical AIE gens, tetraphenylethylene (TPE) was modified in ligands to explore fluorescence properties in the aggregated state.^{5,19,20} To the best of our knowledge, most reported emissive SCCs used gem-TPE with 120° rather *cis*-TPE with 60° motifs as linkers to explore their fluorescence behaviour.^{19,21,22} Compared with gem-TPE, *cis*-TPE motif with a 60° angle can increase ligand rigidity further to decrease the possibility of a mixture of macrocycles. However, it was neglected, which seriously limited the extension of supramolecular architectures and hindered the topology development.

Herein, we designed and constructed two triangular metallo-supramolecules **SA** and **SB** by assembling ditopic terpyridyl ligands (**LA**, **LB**) with *cis*-TPE motif and Zn²⁺. Compared with **LA**, the arms of **LB** were just lengthened with one benzene. The fluorescence properties in dilute solution, aggregation, and solid states of supramolecules were explored. Both **SA** and **SB** showed significant AIE performance in the aggregated state (by increasing diethyl ether fraction) or solid state. It was worth noting that **SA** based on shorter ligands had higher quantum yields than **SB** in both aggregated and solid state. Moreover, triangles **SA** and **SB** displayed tunable emission in different polar solvents.

LA and **LB** were synthesized by Suzuki coupling reaction with yields of 68% and 75%, respectively. The key compound **1** with two boronic acid pinacol ester groups was synthesized according to the reported method.^{23,24} It is worth mentioning that simplified synthetic routes (Scheme 1 and Scheme S1, ESI†) avoided the tedious synthesis and separation of *E/Z* isomers of the precursor **1** with similar polarities *via* McMurry

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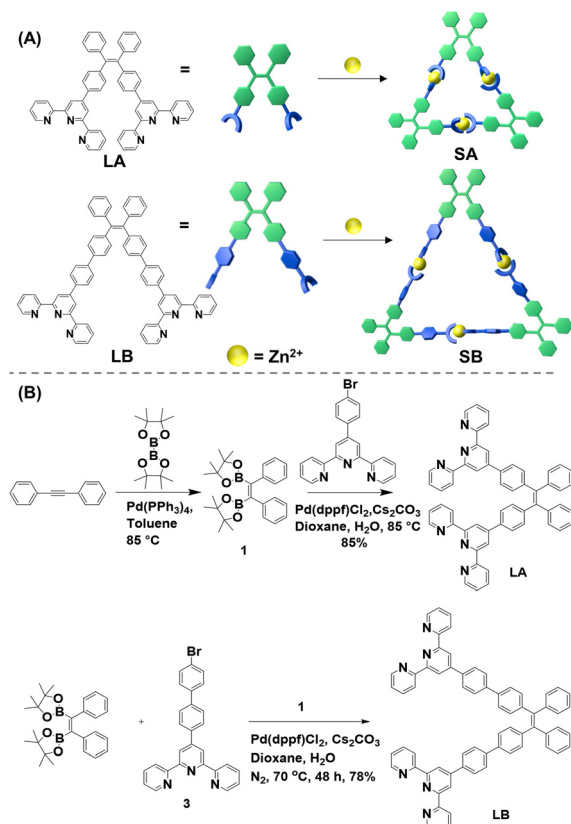
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Scheme 1 (A) Self-assembly of ditopic ligands **LA** and **LB** assembled with Zn^{2+} for constructing metallo-triangles **SA** and **SB**. (B) Synthetic routes of ligands **LA** and **LB**.

coupling reaction and also avoided the follow-up multi-step synthesis of 4'-bromo-2,2',6',2''-terpyridine. The triangles **SA** and **SB** were obtained by mixing ligand **LA** or **LB** and $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ in a 1:1 stoichiometric ratio in a mixture of CHCl_3 and CH_3OH (v:v = 1:1) at 50°C for 12 h (Scheme 1), respectively, and then excessive NH_4PF_6 was added to obtain yellow precipitates in good yields (90% for **SA** and 91% for **SB**, respectively). The structures of **SA** and **SB** were then fully characterized by 1D (^1H and ^{13}C), 2D-COSY NMR spectroscopy, electrospray ionization mass spectrometry (ESI-MS), and traveling wave ion mobility mass spectrometry (TWIM-MS) (Fig. S1–S13, ESI†).

NMR characterization was first applied to address metallo-supramolecular structural information. ^1H NMR spectra of supramolecules **SA** and **SB** and the corresponding ligands **LA** and **LB** are displayed in Fig. 1 and showed sharp and well-splitting signals, which demonstrated the formation of discrete structures. All peaks were accurately assigned by 2D COSY spectra. Compared with ligands **LA** and **LB**, the peaks of $\text{tpy-H}^{3',5'}$ protons of assembled **SA** and **SB** were shifted to downfield ($\Delta\delta = 0.21$ ppm for **SA** and $\Delta\delta = 0.24$ ppm for **SB**) as a result of the lower electron density upon coordination.²⁵ While the $\text{tpy-H}^{6,6''}$ peaks exhibited a significant upfield shift (ca. 0.87 ppm for **SA** and 0.88 ppm for **SB**) as a result of the electron shielding effect, indicating the complexation between tpy and Zn^{2+} .²⁶

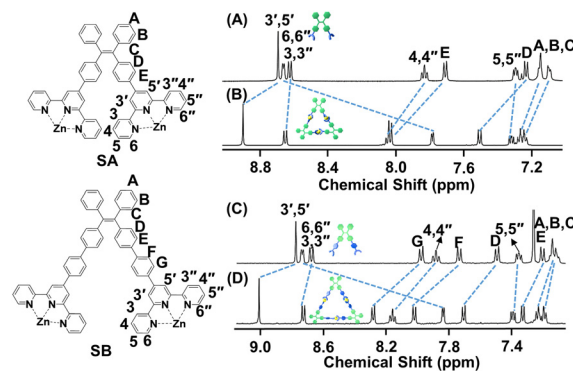


Fig. 1 ^1H NMR spectra (500 MHz, 300 K) of (A) **LA** and (C) **LB** in CDCl_3 and triangles (B) **SA** and (D) **SB** in CD_3CN .

In addition, electrospray ionization-mass spectrometry (ESI-MS) and traveling wave ion mobility-mass spectrometry (TWIM-MS) were utilized to determine the molecular composition of the discrete supramolecules. The mass spectra (Fig. 2A and D) showed one dominant set of peaks with continuous charge states from 3+ to 6+ (due to the loss of a different number of PF_6^-), suggesting discrete supramolecular structures were obtained. After deconvolution of m/z (Fig. 2C and F), the measured molecular masses of the obtained supramolecules were 3444.52 Da and 3900.71 Da, matching well with the molecular compositions of **SA** [$\text{C}_{168}\text{H}_{114}\text{F}_{36}\text{N}_{18}\text{P}_6\text{Zn}_3$] and **SB** [$\text{C}_{204}\text{H}_{138}\text{F}_{36}\text{N}_{18}\text{P}_6\text{Zn}_3$], respectively. The experimental isotope pattern of each charge state also agreed well with the theoretical simulation based on the chemical compositions of **SA** and **SB**. Traveling wave ion mobility-mass spectrum (TWIM-MS) presented a series of signals with narrow bands attributed to **SA** and **SB** with successive charge states, indicating high isomeric purity of the composed structures (Fig. 2B and E).

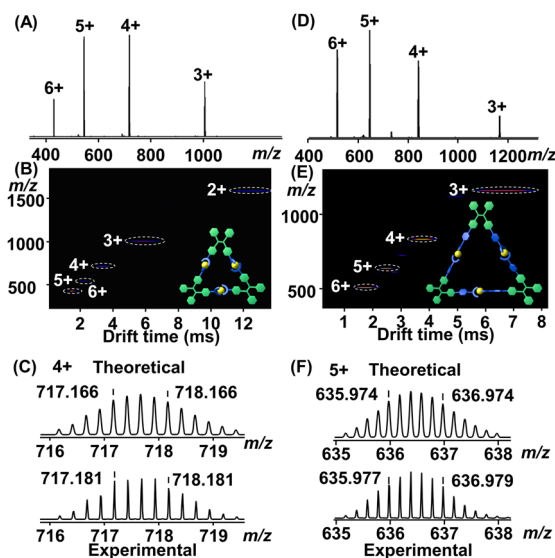


Fig. 2 ESI-MS (A) **SA** and (D) **SB**, TWIM-MS plot (m/z vs. drift time) of (B) **SA** and (E) **SB**, and isotope patterns of (C) **SA** and (F) **SB**. The charge states of the intact assemblies are marked.

Encouraged by the successful construction of two discrete structures **SA** and **SB**, the fluorescence properties of ligands and supramolecules in dilute solution were investigated. **LA** and **LB** in chloroform exhibited main absorption peaks among 250–300 nm, attributing to the π - π^* transition of tpy and conjugated core TPE (Fig. 3A).²⁷ The absorption band in the range of 350–400 nm was ascribed to the intramolecular charge transfer (ICT) absorption from an electron donor (TPE) to the electron acceptor (tpy) moieties. Fluorescence spectra of ligands **LA** and **LB** displayed weak emission at about 400 and 520 nm in chloroform (absolute fluorescence quantum yield $\Phi_F = 1.3\%$ for **LA** and $\Phi_F = 0.3\%$ for **LB**) owing to the non-radiative decay resulting from the free rotation of phenyl groups. Compared with these two ligands, the main absorption bands of metallo-triangles **SA** and **SB** (Fig. 3B) were bathochromically shifted by 30 and 80 nm (284 and 330 nm). As shown in Fig. 3B, fluorescence emission spectra of complexes **SA** and **SB** underwent a moderate redshift, which could result from a specific conformation of the peripheral phenyl rings: a coplanar conformation promotes π -electron conjugation after coordination.²⁸ The quantum yields of **SA** (1.27%) and **SB** (1.24%) in dilute solution ($c = 10^{-6}$ M) have unobvious fluorescence enhancement, which suggested that the non-radiative decay pathways of TPE motifs were not limited effectively.

To further explore the AIE properties, the fluorescent emission of **SA** and **SB** in a mixed solution of CH_3CN ($c = 10^{-6}$ M) as good solvent and diethyl ether as poor solvent were investigated. The fluorescence intensity and quantum yields of **SA** and **SB** showed no obvious change when diethyl ether content is below 70% (Fig. 4A–C) due to the good solubility of these two metallo-triangles in the mixed solution. However, compared with low diethyl ether fractions, the fluorescence intensity of **SA** and **SB** increased sharply with increasing diethyl ether content up to 90%, with $\Phi_F = 28.18\%$ and $\Phi_F = 10.88\%$ (Fig. 4C), respectively. The quantum yields were 22.2-fold and 8.7-fold than its dispersed species in CH_3CN . All the collected results indicated the obtained metallo-triangles with *cis*-TPE motifs had significant AIE-active features. Moreover, **SA** showed higher quantum yield than **SB** at 90% diethyl ether content, which may be because the rigid and compact skeleton **SA** and distributed distance of AIE restricted the rotation of phenyl rings and blocked non-radiative decay pathway. However, different from the accompanying slight blueshift of the fluorescent spectrum of **SA** with increasing diethyl ether fraction, **SB** displayed a bathochromic shift. The possible reasons are as below: (i) charge-transfer (CT) processes could redshift the fluorescence colour and

decrease the emission efficiency. (ii) Less-polar solvent attenuated charge-transfer (CT) processes, and thereby causing a blue shift in emission and an increase in emission efficiency.²⁸ The balance between CT processes and less-polar solvent should be considered and also consistent with the observed difference in emission shifts between **SA** and **SB**. Overall, by increasing the fraction of poor solvent, the rotation of phenyl rings was restricted by aggregation, leading to AIE behaviour.

The fluorescence properties of triangles **SA** and **SB** were further evaluated in different polar solvents at 10^{-5} M. These two triangles **SA** and **SB** (Fig. 4D and E) showed apparent changes in intensity and shifts with irregular changes with increasing polarity of solvents, which indicated that polarity of the solvent has an underestimated effect on aggregation, which in turn affected the electronic structure of supramolecules.²⁹ Triangles **SA** and **SB** had pronounced colours (orange and green) and intensity variations according to different polar solvents, indicating that solvent polarity would affect emission behaviour. The quantum yields in different solvents in dilute solution are lower than in high diethyl ether fractions in diethyl ether/ CH_3CN (Table S2, ESI†). In moderate-polarity solvents, including acetone, CH_3NO_2 , and CH_3CN , orange colour with subtle changes was observed, whereas green colour could appear in high-polar solvent, such as DMSO.

Except for optical properties in solution, fluorescent properties of ligands and supramolecules in the solid state were also recorded. From Fig. S14 (ESI†), **LA** was bathochromically shifted *ca.* 50 nm in the fluorescent spectrum, in comparison with **LB**, which should be a CT process. In contrast with quantum yields (1.27% and 1.24%) in dilute acetonitrile solution (10^{-6} M), **SA** and **SB** showed high quantum yields $\Phi_F = 18.35\%$ and 7.29% in solid states (Fig. 4F), corresponding to 14.4 fold and 5.87 fold, respectively, which resulted from the restriction of intramolecular rotation of phenyl rings after aggregation, which fits the classical AIE phenomenon. Compared with the quantum yields of high diethyl ether fraction in dilute solution, the quantum yields in the solid state were lower, which may be caused by excessive molecular packing. Moreover, compared with yellow **SA**, triangle **SB** with the orange colour displayed redshift and lower quantum yield on account of increasing arm lengths of ligands, causing larger molecular dipole moment.³⁰

In summary, two *cis*-TPE-containing metallo-supramolecular triangles—short version **SA** and long version **SB** were designed and constructed by assembling ditopic ligands with different arm lengths and Zn^{2+} , respectively. **SA** and **SB** were identified by ^1H NMR, ^{13}C NMR, COSY, ESI-MS, and TWIM-MS. Apart from the structural characterization, the optical properties of two triangles have been investigated in different aggregation states and polar solvents. Both triangles displayed classical AIE features with high quantum yields in the aggregation process, including increasing diethyl ether fraction in dilute solution and solid states. It is worth noting that **SA** showed yellow-green emission, and **SB** displayed pronounced orange emission in the aggregation state. Moreover, the two triangles displayed tunable emission in different polar solvents. This work opens a new perspective and opportunity to study the construction of metallosupramolecules and luminescent materials.

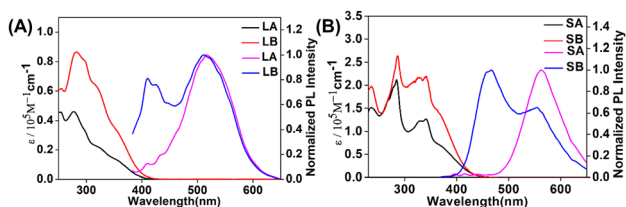


Fig. 3 (A) UV-vis and PL spectra of **LA** and **LB** in CHCl_3 ($\lambda_{\text{ex}} = 340$ nm, $c = 10^{-5}$ M). (B) UV-vis and PL spectra of **SA** and **SB** in acetonitrile ($\lambda_{\text{ex}} = 340$ nm, $c = 10^{-5}$ M).

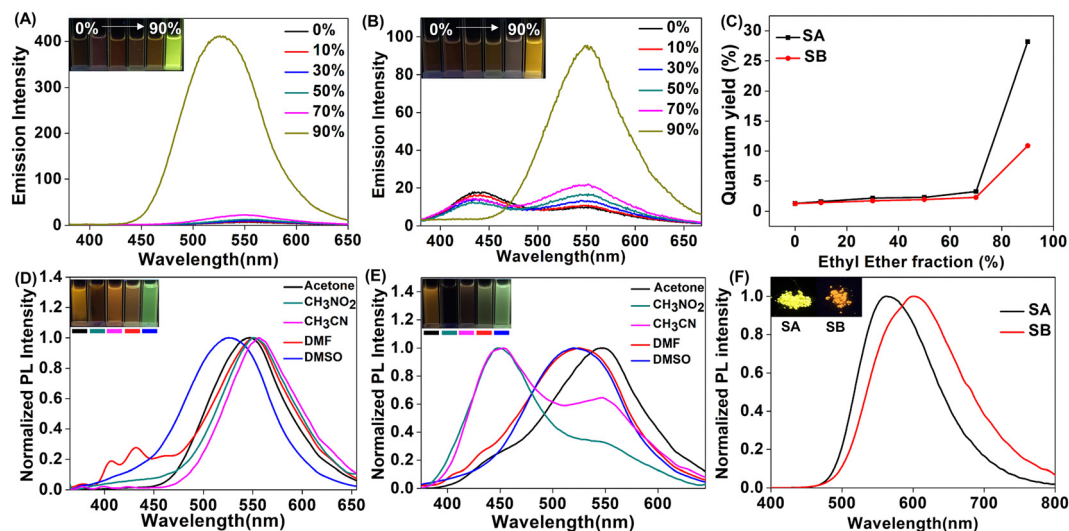


Fig. 4 PL spectra of (A) SA and (B) SB versus diethyl ether in CH₃CN/diethyl ether mixtures (λ_{ex} = 340 nm, c = 10^{-6} M) with insert photographs of different diethyl ether fractions excited by 365 nm UV lamp at 300 K; quantum yields (C) of SA and SB; solvent effect on the light-emitting properties of (D) SA and (E) SB (λ_{ex} = 340 nm, c = 10^{-5} M) with insert photographs of different solvents excited by a 365 nm UV lamp at 300 K; normalized PL spectra (F) of SA and SB in the solid state (λ_{ex} = 340 nm) with insert photographs of SA and SB in the solid state excited by 365 nm UV lamp at 300 K. CH₃NO₂, nitromethane; CH₃CN, acetonitrile; DMF, *N,N*-dimethylformamide; DMSO, dimethyl sulfoxide.

Junjuan Shi: idea conception, designing the synthetic routes, synthesis of ligands, self-assembly process, analysis of all data including NMR, ESI-MS, TWIM-MS, UV-vis, and fluorescence spectra, and manuscript writing. Meng Li: collection UV-vis and fluorescence spectra. Haoyue Su, Ningxu Han, Xin-Qi Hao: synthesis of ligands and self-assembly process. Qixia Bai, Zhe Zhang, Fang Fang and Pingshan Wang: collection of ESI-MS and TWIM-MS spectra. Benhua Ma and Ming Wang: supervision, project administration, writing-review, and editing. All authors have discussed and given approval for the final version of the manuscript.

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Conflicts of interest

There are no conflicts to declare.

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